QBIO 490: Directed Research - Multi-Omic Analysis
Methylation Homework [Fall 2026]

Due: Monday, November 2, 2026 (4:59 pm). Submit your GitHub link to Brightspace, with your R Notebook with all your code and code outputs in the `week08_methylation` folder within your `qbio_490_name` repo. Please email extension requests (include the reason for your extension and a proposed new due date) to Josh and Chris by November 1st.

Be sure to review the supplemental video and notebook when starting this homework: https://drive.google.com/file/d/10guitWgrBwpwxKpbO58osuyr0TgyvhqX/view?usp=drive_link

Purpose: What are the differences between metastatic and non-metastatic BRCA across the epigenome?

Exploration of Methylation Patterns: Perform each analysis in a new R Markdown file. For each, write a 3-4 sentence description analyzing the plot and answering the following question: What conclusions can you and can you not draw about differences between metastatic and non-metastatic TCGA BRCA patients? Why?

1. **Naive differential methylation between metastatic and non-metastatic patients (Volcano plot):**
    a. Refer to Part 1 of the video/demo: instead of age_at_diagnosis (old/young), consider the methylation_clinical@colData$definition column to subset your groups.

2. **Direct comparison of methylation status to transcriptional activity across non-metastatic vs metastatic patients in 1 gene of interest (Box plot & bar plot):**
    a.
```r
#Extract genes of interest
downregulated <- de_results[(de_results$log2FoldChange < -1), 'gene_name']
hypermethylated <- dat[dat$foldchange > 1.5, 'geneName'] #what cutoff should you use here?
interest_genes <- intersect(downregulated, hypermethylated)
```
    b. Workflow:
        i. Bring back your DESeq ‘results’ table
        ii. (like the picture above) extract:
            1. from DESeq, downregulated gene_names
            2. From dat (methylation), hypermethylated gene_names
        iii. Intersect these two lists of gene_names, and pick one.
        iv. Use selected gene to plot box/bar plots.

3. **[EXTRA CREDIT, +1 per gene]** Visualization of CpG sites and protein domains for **MAX 3 genes** (use UCSC genome browser) for a few genes. Describe at least one academic article (research or review) that either supports or doesn't support your final conclusion for one of the genes. If previously published work doesn't support your analysis, explain why this might be the case.

Technical Tips:

- Metastasis status should be based on the methylation_clinical@colData$definition column.
    - Only consider “Metastatic” or “Primary solid Tumor” samples
- Be careful about what “barcode” columns you use! The patient id, sample id, and sample barcode columns are all named slightly differently across the different dataframes. Double check that the columns you are using to match index values are correct!
- For DESeq2 data preprocessing:
    - Use the methylation clinical data (methylation_clinical@colData).
    - Filter out genes with a total expression across all patients of < 20
    - Threshold padj values at 0.05 and log2FoldChange at |1|
- UCSC settings
    - GENCODE V49: dense
    - MANE: pack
    - UniProt: dense
    - EPDnewPromoters: dense
    - CpG Islands: show